

Lemma G.8. *There exist constants $c_0, \rho_0 \in (0, 1)$, depending only on A , such that for every $0 < \rho \leq \rho_0$, every $t \geq 0$, and every $\mathbf{u} \in \mathbb{S}^{d-1}$ satisfying $\|\mathbf{u} - \boldsymbol{\mu}_t\| \leq \frac{1}{4}$,*

$$\|\Pi((1 - \rho)\mathbf{u} + \rho A \boldsymbol{\mu}_t) - \boldsymbol{\mu}_t\| \leq (1 - c_0 \rho) \|\mathbf{u} - \boldsymbol{\mu}_t\|.$$

Proof of Lemma G.8. Fix t and $\mathbf{u} \in \mathbb{S}^{d-1}$ with $\|\mathbf{u} - \boldsymbol{\mu}_t\| \leq 1/4$. Write

$$\mathbf{u} = \alpha \boldsymbol{\mu}_t + \beta \boldsymbol{\nu}, \quad \boldsymbol{\nu} \perp \boldsymbol{\mu}_t, \quad \alpha^2 + \beta^2 = 1.$$

Since $\|\mathbf{u} - \boldsymbol{\mu}_t\|^2 = 2(1 - \alpha)$, the condition $\|\mathbf{u} - \boldsymbol{\mu}_t\| \leq 1/4$ implies $\alpha \geq 1 - \frac{1}{32} = \frac{31}{32}$. Define

$$\mathbf{x} := (1 - \rho)\mathbf{u} + \rho A \boldsymbol{\mu}_t = a \boldsymbol{\mu}_t + b \boldsymbol{\nu},$$

where $a := (1 - \rho)\alpha + \rho A$ and $b := (1 - \rho)\beta$. Then

$$\Pi(\mathbf{x}) = \frac{a \boldsymbol{\mu}_t + b \boldsymbol{\nu}}{r} \quad \text{with} \quad r := \sqrt{a^2 + b^2}.$$

Therefore, we have that

$$\|\Pi(\mathbf{x}) - \boldsymbol{\mu}_t\|^2 = 2 \left(1 - \frac{a}{r}\right) = \frac{2b^2}{r(r + a)}.$$

Using the facts that $\|\mathbf{u} - \boldsymbol{\mu}_t\|^2 = 2(1 - \alpha)$ and $\beta^2 = 1 - \alpha^2$, we get

$$\frac{\|\Pi(\mathbf{x}) - \boldsymbol{\mu}_t\|^2}{\|\mathbf{u} - \boldsymbol{\mu}_t\|^2} = \frac{(1 - \rho)^2(1 + \alpha)}{r(r + a)} =: \Psi(\alpha, \rho).$$

Now $\alpha \in [31/32, 1]$, and one checks that $r^2 = (1 - \rho)^2 + 2\rho(1 - \rho)A\alpha + \rho^2 A^2$ by definition. Hence, Ψ is continuous on the compact set $[31/32, 1] \times [0, \rho_1]$ for any fixed $\rho_1 < 1$, and $\Psi(\alpha, 0) = 1$ for all α . Moreover,

$$\left. \frac{\partial}{\partial \rho} \Psi(\alpha, \rho) \right|_{\rho=0} = -A(1 + \alpha) \leq -A \cdot \frac{63}{32} < 0,$$

uniformly over $\alpha \in [31/32, 1]$. Therefore, by compactness and continuity, there exist constants $c_0, \rho_0 \in (0, 1)$, depending only on A , such that for all $\alpha \in [31/32, 1]$ and all $0 < \rho \leq \rho_0$,

$$\Psi(\alpha, \rho) \leq (1 - c_0 \rho)^2.$$

Taking square roots yields that

$$\|\Pi((1 - \rho)\mathbf{u} + \rho A \boldsymbol{\mu}_t) - \boldsymbol{\mu}_t\| \leq (1 - c_0 \rho) \|\mathbf{u} - \boldsymbol{\mu}_t\|.$$

□

Lemma G.9 (Weighted martingale concentration). *Let $\{\Delta_{t+1}\}_{t \geq 0}$ be a martingale difference sequence with respect to $\{\mathcal{F}_t\}$. Assume that there exist constants $B, V > 0$ such that, almost surely,*

$$|\Delta_{t+1}| \leq B\rho, \quad \mathbb{E}[\Delta_{t+1}^2 | \mathcal{F}_t] \leq V\rho^2 \quad \forall t \geq 0.$$

Fix $c_*, \rho \in (0, 1)$ and define, for each $t \geq 1$,

$$S_t := \sum_{s=0}^{t-1} (1 - c_* \rho)^{t-1-s} \Delta_{s+1}.$$

Then there exists a constant $C > 0$, depending only on B, V, c_* , such that for any $\delta \in (0, 1)$,

$$\mathbb{P} \left(\sup_{1 \leq t \leq T} |S_t| \leq C \left(\sqrt{\rho \log \frac{T}{\delta}} + \rho \log \frac{T}{\delta} \right) \right) \geq 1 - \delta.$$

Proof of Lemma G.9. Fix $t \geq 1$. Define

$$\xi_{t,s+1} := (1 - c_* \rho)^{t-1-s} \Delta_{s+1}, \quad 0 \leq s \leq t-1.$$

Then $\{\xi_{t,s+1}\}_{s=0}^{t-1}$ is a martingale difference sequence with respect to $\{\mathcal{F}_s\}$, and

$$|\xi_{t,s+1}| \leq (1 - c_* \rho)^{t-1-s} B\rho.$$

Moreover,

$$\sum_{s=0}^{t-1} \mathbb{E}[\xi_{t,s+1}^2 \mid \mathcal{F}_s] \leq V\rho^2 \sum_{s=0}^{t-1} (1 - c_\star \rho)^{2(t-1-s)} \leq V\rho^2 \sum_{j=0}^{\infty} (1 - c_\star \rho)^{2j}.$$

Since

$$\sum_{j=0}^{\infty} (1 - c_\star \rho)^{2j} = \frac{1}{1 - (1 - c_\star \rho)^2} \leq \frac{1}{c_\star \rho},$$

it follows that

$$\sum_{s=0}^{t-1} \mathbb{E}[\xi_{t,s+1}^2 \mid \mathcal{F}_s] \leq \frac{V}{c_\star} \rho.$$

Applying Freedman's inequality [84] yields that, for any $u > 0$,

$$\mathbb{P}(|S_t| \geq u) \leq 2 \exp\left(-\frac{u^2}{2\left(\frac{V}{c_\star} \rho + \frac{1}{3} B \rho u\right)}\right).$$

Taking a union bound over $1 \leq t \leq T$ and choosing

$$u = C\left(\sqrt{\rho \log \frac{T}{\delta}} + \rho \log \frac{T}{\delta}\right).$$

for a sufficiently large constant C proves the claim. $\square$

H Additional Theory

H.1 Gradient Analysis of the Population Training Objective

The training procedure in Phase I follows a two-timescale scheme: the steering vector $\mathbf{v}$ is updated on a slow timescale, while the class mean directions are updated on a fast timescale. To make the role of this dynamics clear, we analyze the corresponding population objective.

For a fixed steering vector $\mathbf{v}$, we then define

$$\mathcal{J}(\mathbf{v}) := \max_{\boldsymbol{\mu}_0, \boldsymbol{\mu}_1 \in \mathbb{S}^{d-1}} \mathbb{E}_{x,y}[\log q_\beta(y \mid f_{\theta,\mathbf{v}}(x), \boldsymbol{\mu}_0, \boldsymbol{\mu}_1)] = \mathbb{E}_{x,y}[\log q_\beta(y \mid f_{\theta,\mathbf{v}}(x), \boldsymbol{\mu}_0(\mathbf{v}), \boldsymbol{\mu}_1(\mathbf{v}))].$$

where $\boldsymbol{\mu}_c(\mathbf{v})$ denotes the population-optimal mean direction for class c induced by the steered representation map $f_{\theta,\mathbf{v}}$.

The following proposition characterizes how optimizing $\mathcal{J}(\mathbf{v})$ reshapes the latent geometry. For this analysis, we work under Assumption 1, a uniform class prior $p(y=0) = p(y=1) = 1/2$, and the matched case $\beta = \kappa$. Under these conditions, the temperature-scaled posterior q_β in (2) coincides with the class posterior induced by the shared-concentration vMF model.

Proposition H.1. *The gradient of $\mathcal{J}(\mathbf{v})$ admits the decomposition*

$$\nabla_{\mathbf{v}} \mathcal{J}(\mathbf{v}) = \beta \bar{\omega}(\mathbf{v}) \nabla_{\mathbf{v}} \|\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})\|^2 + \frac{\beta}{2} (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\Delta_1(\mathbf{v}) - \Delta_0(\mathbf{v})).$$

Here, $\Delta_0(\mathbf{v})$ and $\Delta_1(\mathbf{v})$ are residual terms defined in (23) as

$$\Delta_c(\mathbf{v}) := \mathbb{E}_x[q_\beta(1 - c \mid x)(\nabla_{\mathbf{v}} f_{\theta,\mathbf{v}}(x) - \nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v})) \mid y = c], \quad (23)$$

and $\bar{\omega}(\mathbf{v}) \geq 0$ is a data-dependent weight given by

$$\bar{\omega}(\mathbf{v}) = \frac{1}{8} \left(1 - \mathbb{E}_x \left[\tanh^2 \left(\frac{\beta}{2} (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top f_{\theta,\mathbf{v}}(x) \right) \right] \right).$$

Proposition H.1 decomposes the gradient of the population objective into two components. The first term drives an increase in the separation between the class mean directions $\|\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})\|^2$, with a data-dependent weight that emphasizes samples near the decision boundary. The second term

captures a residual effect arising from the mismatch between the sample-level response $\nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x)$ and the induced movement of the class means.

This decomposition clarifies the optimization dynamics: when the residual term is small, the first term pushes the representation toward greater class separation, while the residual term captures a correction that depends on local variability.

Proof of Proposition H.1. Denote by

$$\mathcal{L}_{\beta}(\mathbf{v}, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1) := \log q_{\beta}(y \mid f_{\theta, \mathbf{v}}(x), \boldsymbol{\mu}_0, \boldsymbol{\mu}_1).$$

Then $\mathcal{J}(\mathbf{v}) = \mathbb{E}_{x,y}[\mathcal{L}_{\beta}(\mathbf{v}, \boldsymbol{\mu}_0(\mathbf{v}), \boldsymbol{\mu}_1(\mathbf{v}))]$. Differentiating with respect to $\mathbf{v}$ gives

$$\nabla_{\mathbf{v}} \mathcal{J}(\mathbf{v}) = \mathbb{E}_{x,y}[\nabla_f \mathcal{L}_{\beta}^{\top} \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x)] + \sum_{c \in \{0,1\}} \left(\nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v}) \right)^{\top} \nabla_{\boldsymbol{\mu}_c} \mathcal{J}(\mathbf{v}).$$

Since $\boldsymbol{\mu}_c(\mathbf{v})$ is the optimal class mean direction on $\mathbb{S}^{d-1}$, the first-order optimality condition implies that $\nabla_{\boldsymbol{\mu}_c} \mathcal{J}(\mathbf{v})$ is normal to $\mathbb{S}^{d-1}$ at $\boldsymbol{\mu}_c(\mathbf{v})$. On the other hand, $\|\boldsymbol{\mu}_c(\mathbf{v})\| = 1$ for all $\mathbf{v}$, and differentiating $\|\boldsymbol{\mu}_c(\mathbf{v})\|^2 = 1$ yields $\boldsymbol{\mu}_c(\mathbf{v})^{\top} \nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v}) = 0$, which shows that $\nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v})$ lies in the tangent space of $\mathbb{S}^{d-1}$ at $\boldsymbol{\mu}_c(\mathbf{v})$. Consequently,

$$\left(\nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v}) \right)^{\top} \nabla_{\boldsymbol{\mu}_c} \mathcal{J}(\mathbf{v}) = 0,$$

which implies that

$$\nabla_{\mathbf{v}} \mathcal{J}(\mathbf{v}) = \mathbb{E}_{x,y} \left[(\nabla_f \mathcal{L}_{\beta})^{\top} \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) \right]. \quad (24)$$

For simplicity, write $q_{\beta}(y \mid x) := q_{\beta}(y \mid f_{\theta, \mathbf{v}}(x), \boldsymbol{\mu}_0(\mathbf{v}), \boldsymbol{\mu}_1(\mathbf{v}))$. Under the matched case $\beta = \kappa$, this coincides with the class posterior induced by the shared-concentration vMF model, and

$$q_{\beta}(y \mid x) = \frac{\exp\{\beta \boldsymbol{\mu}_y(\mathbf{v})^{\top} f_{\theta, \mathbf{v}}(x)\}}{\exp\{\beta \boldsymbol{\mu}_0(\mathbf{v})^{\top} f_{\theta, \mathbf{v}}(x)\} + \exp\{\beta \boldsymbol{\mu}_1(\mathbf{v})^{\top} f_{\theta, \mathbf{v}}(x)\}}.$$

A direct calculation gives

$$\nabla_f \mathcal{L}_{\beta} = \beta \left(\boldsymbol{\mu}_y(\mathbf{v}) - q_{\beta}(0 \mid x) \boldsymbol{\mu}_0(\mathbf{v}) - q_{\beta}(1 \mid x) \boldsymbol{\mu}_1(\mathbf{v}) \right),$$

which can be rewritten as

$$\nabla_f \mathcal{L}_{\beta} = \beta (y - q_{\beta}(1 \mid x)) (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})).$$

Substituting this into (24) and conditioning on y yields

$$\begin{aligned} \nabla_{\mathbf{v}} \mathcal{J}(\mathbf{v}) = \frac{\beta}{2} (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^{\top} & \left(\mathbb{E}_x[q_{\beta}(0 \mid x) \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) \mid y = 1] \right. \\ & \left. - \mathbb{E}_x[q_{\beta}(1 \mid x) \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) \mid y = 0] \right), \end{aligned} \quad (25)$$

where we used $p(y = 0) = p(y = 1) = 1/2$. For each $c \in \{0, 1\}$, introduce the residual term

$$\Delta_c(\mathbf{v}) := \mathbb{E}_x[q_{\beta}(1 - c \mid x) (\nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) - \nabla_{\mathbf{v}} \boldsymbol{\mu}_c(\mathbf{v})) \mid y = c]. \quad (23)$$

Then, we obtain

$$\mathbb{E}_x[q_{\beta}(0 \mid x) \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) \mid y = 1] = \mathbb{E}_x[q_{\beta}(0 \mid x) \mid y = 1] \nabla_{\mathbf{v}} \boldsymbol{\mu}_1(\mathbf{v}) + \Delta_1(\mathbf{v}),$$

and

$$\mathbb{E}_x[q_{\beta}(1 \mid x) \nabla_{\mathbf{v}} f_{\theta, \mathbf{v}}(x) \mid y = 0] = \mathbb{E}_x[q_{\beta}(1 \mid x) \mid y = 0] \nabla_{\mathbf{v}} \boldsymbol{\mu}_0(\mathbf{v}) + \Delta_0(\mathbf{v}).$$

Substituting these identities into (25) gives

$$\begin{aligned} \nabla_{\mathbf{v}} \mathcal{J}(\mathbf{v}) = \frac{\beta}{2} (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^{\top} & \left(\mathbb{E}_x[q_{\beta}(0 \mid x) \mid y = 1] \nabla_{\mathbf{v}} \boldsymbol{\mu}_1(\mathbf{v}) \right. \\ & \left. - \mathbb{E}_x[q_{\beta}(1 \mid x) \mid y = 0] \nabla_{\mathbf{v}} \boldsymbol{\mu}_0(\mathbf{v}) \right) \\ & + \frac{\beta}{2} (\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^{\top} (\Delta_1(\mathbf{v}) - \Delta_0(\mathbf{v})). \end{aligned} \quad (26)$$